

11/7/2026 Short answer

Kishore Kumar

192624108

DSA0405

1. What is Data Science? Explain its Primary Objective.

Data Science is a interdisciplinary field that combines statistics, Programming, machine Learning and domain knowledge to extract meaningful insights from data. It involves collecting, processing, analyzing and interpreting large amount of data.

2. Why is Data Science important in today digital world? mention any four reasons.

Data Science plays an important role in today's digital world. Because huge amounts of data are generated every day through social media, online transactions and smart devices.

3. List any four Benefits of Data Science in Business and Society.

Data Science provides numerous benefits to business and society. It helps companies understand customer behaviour, improve productivity, reduce costs, and make informed decisions.

q4.1.

28/30

11/7/26



4. Mention four real-world application of Data Science

Data Science has many real-world applications across different industries. It is used in recommendation systems such as Netflix and Amazon, fraud detection in banking, disease prediction in healthcare.

5. What are the different facets of data? Briefly explain each.

The main facets of data are structure, source, type, time, quality and usage. Structure refers to the organization of data, source indicates where the data comes, and type describes the nature of data values.

6. Differentiate between structured, semi-structured and unstructured data

Structured data is organized in row and columns, such as database. Semi-structured data contains tags, JSON and XML. Unstructured data include image, video and emails.

7. What is Big Data? State its key characteristics.

Big Data refers to very large and complex datasets. Its five characteristics are volume, velocity, variety, veracity and value.



8. Explain the Big Data ecosystem. Name any four technologies used in it.

The Big Data ecosystem consists of tools used to store, process, and analyze large datasets. Some technologies are Hadoop, Spark

9. What are the major phases of the Data Science Process

The major phases are data collection, data cleaning, data integration, data transformation, data analysis, and data visualization

10. What is data retrieval? Mention two common sources

Data retrieval is the process of obtaining data from storage systems. Common sources include data bases and websites



11. why is data collection considered important in Data Science?

Data collection is important because accurate and relevant data leads to reliable analysis and better decision-making.

12. what is data cleansing? list any four techniques.

Data cleansing is the process of correcting error in data. Common techniques include removing duplicates, handling missing values.

13. what is data integration? why is it necessary?

Data integration combines data from different sources into a single dataset.

14. Explain data transformation with suitable example.

Data transformation converts data into a suitable format for analysis.

15. what is data analysis? mention any four techniques?

Data analysis is the process of examining data to extract useful information.